

Department of Physics
University of Florida

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PHY 2049

Final Exam

April 29, 2025

On my honor, I have neither given nor received unauthorized aid on this examination

Name _____

Last: _____ First: _____ Signature: _____

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Directions

1. Print your name on this question sheet and sign it.
2. Code your 5-digit exam code (at the top of this question sheet) on your Scantron in columns 76–80.
3. Code your name and 8-digit UFID on the Scantron. **Darken all circles completely on the Scantron** using a #2 pencil or **dark blue** or **black** ink. Do not make any stray marks or the Scantron may not read properly.
4. Do all scratch work anywhere on this question sheet that you like. At the end of the exam, the question sheet and Scantron are both to be turned in. *No credit will be given without both Scantron sheet and question sheet.*
5. The answers are rounded, so choose the closest to exact. There is no penalty for guessing.

Correct answer marked with ♣

1. Three point charges of $200\ \mu\text{C}$ each are positioned in the (x,y) -plane at points A(-3,0), B(0,4), C(3,0), respectively. The coordinates are in meters. What is the magnitude of the net force on the charge placed at point C?

- (1) ♣ 21.9 N
 (2) 28.8 N
 (3) 17.3 N
 (4) 33.3 N
 (5) 24.4 N

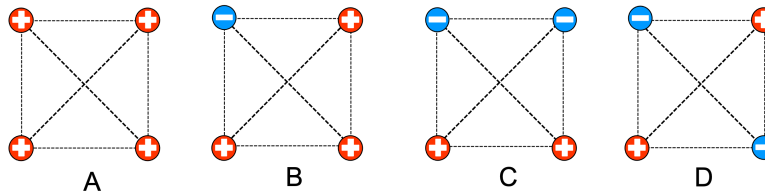
Solution: For this problem you need to draw a force diagram and compute the x and y components of the force on C separately.

$$F_x = \frac{kq^2}{6^2} + \frac{kq^2}{5^2} \frac{3}{5} = 18.64\ \text{N}$$

$$F_y = -\frac{kq^2}{5^2} \frac{4}{5} = -11.52\ \text{N}$$

The factors of $3/5$ and $4/5$ are the cosine and sine factors for the x and y components of the force on C due to B. The magnitude of the force is $\sqrt{F_x^2 + F_y^2} = \mathbf{22.9\ \text{N}}$.

2. Which of the configurations with four charges placed at the corners of squares, as shown below, has the strongest electric field at the square's center? The plus symbol indicates a positive charge $+q$, and the minus symbol indicates a negative charge $-q$.

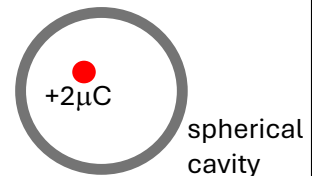


- (1) ♣ C (2) A (3) B (4) D (5) insufficient information

Solution: When the charges on diagonally opposite corners have the same charge, the electric fields are in opposite directions and cancel. This occurs for both diagonal pairs in A and D so the net electric field at the center in those cases is zero. For case B one diagonal pair has electric fields which cancel, while for case C neither of them cancel. Thus, the electric field is largest for case C.

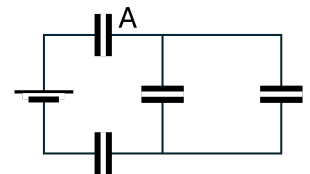
3. A small ball, with a charge of $+2\ \mu\text{C}$ on it, is placed in a spherical cavity inside a spherical conductor, as illustrated in the figure. The net charge on the conductor is $+2\ \mu\text{C}$. What is the amount of charge on the *outer* surface of the conductor?

- (1) ♣ $+4\ \mu\text{C}$
 (2) $+2\ \mu\text{C}$
 (3) $0\ \mu\text{C}$
 (4) $-4\ \mu\text{C}$
 (5) $-2\ \mu\text{C}$



Solution: Consider a Gaussian surface inside the conducting shell. It has zero flux because the electric field inside a conductor is zero. Thus, there must be no net charge inside the Gaussian surface, which means the charge on the inside surface of the conducting shell is $-2\ \mu\text{C}$. Since the net charge on the conductor is $+2\ \mu\text{C}$, the outside surface must have charge $+4\ \mu\text{C}$.

4. In the capacitor circuit in the figure all the capacitors are $3\ \mu\text{F}$, and the battery is 6 V. What is the energy stored in the capacitor marked "A"?



- (1) ♣ $8.6\ \mu\text{J}$ (2) $54\ \mu\text{J}$ (3) $21.6\ \mu\text{J}$ (4) $2.2\ \mu\text{J}$ (5) $216\ \mu\text{J}$

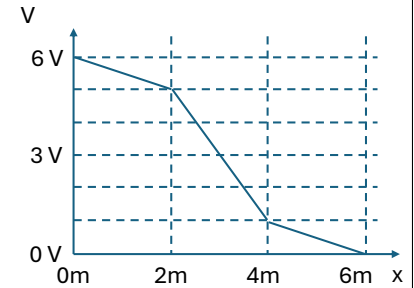
Solution: First compute the equivalent capacitance of the entire circuit. The two capacitors on the right are in parallel. They have an equivalent capacitance of $6\ \mu\text{F}$. These capacitors are then in series with the other two capacitors.

$$\frac{1}{C_{eq}} = \frac{1}{3\mu\text{F}} + \frac{1}{6\mu\text{F}} + \frac{1}{3\mu\text{F}} \rightarrow C_{eq} = 1.2\mu\text{F}.$$

The charge on this equivalent capacitor is $(6\text{V})(1.2\mu\text{F}) = 7.2\mu\text{C}$. This is also the charge on the capacitor marked as “A” because capacitors in series have the same charge. The energy stored in capacitor “A” is $q^2/2C = 0.5(7.2\mu\text{C})^2/(3\mu\text{F}) = 8.6\mu\text{J}$.

5. In the figure the voltage as a function of position is shown for three wires connected in series. The segment between $x = 2\text{ m}$ and $x = 4\text{ m}$ has resistivity $2 \times 10^{-6}\ \Omega\text{m}$ and diameter 0.5 mm . What is the current density in the wire?

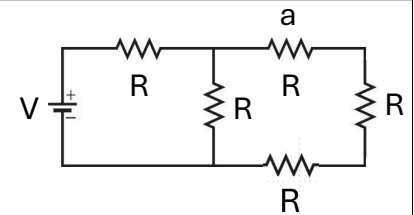
- (1) ♣ 1 A/mm^2
- (2) 2 A/mm^2
- (3) 3 A/mm^2
- (4) 4 A/mm^2
- (5) 0.3 A/mm^2



Solution: The current density is $j = (1/\rho)E$. The electric field between 2 m and 4 m is $4\text{V}/2\text{m} = 2\text{ V/m}$. Thus, the current density is $2/2 \times 10^{-6} = 10^{-6}\text{ A/m}^2 = 1\text{ A/mm}^2$.

6. 1. In the resistor circuit shown in the figure each of the resistors has resistance $2\ \Omega$, and the battery has voltage 7 V . What is the voltage across the resistor labelled as “a”?

- (1) ♣ 1 V
- (2) 2 V
- (3) 3 V
- (4) 4 V
- (5) 5 V

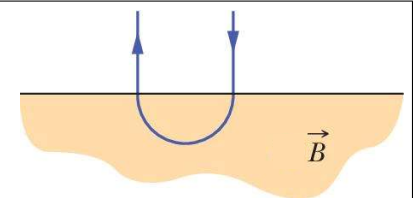


Solution: First compute the equivalent resistance of the entire circuit. The three resistors on the right are in series and have equivalent resistance of $6\ \Omega$. These three resistors are in parallel with the middle resistor creating an equivalent resistance of $1/R_{eq} = 1/6 + 1/2 = 2/3$ or $R_{eq} = 1.5\ \Omega$. This is in series with the remaining resistor so the equivalent resistance of the entire circuit is $3.5\ \Omega$.

The current through the entire circuit is $7/3.5 = 2\text{ A}$. The voltage drop across the left resistor is $(2\Omega)(2\text{A}) = 4\text{V}$. This means the voltage drop across the remaining resistors is $7\text{V} - 4\text{V} = 3\text{V}$. This is the voltage drop across the central resistor and also the three resistors on the right. Since the current through the three resistors on the right is the same, the voltage drop across each resistor is the same, and the voltage drop across resistor “a” is $3\text{V}/3 = 1\text{V}$.

7. An electron enters a region with a uniform magnetic field and follows a semicircular path, which is half of a complete orbit. It spends 3 ns in the magnetic field. What is the magnitude and direction of the magnetic field? The mass of an electron is $9.1 \times 10^{-31}\text{ kg}$.

- (1) ♣ 6 mT , into the page
- (2) 6 mT , out of the page
- (3) 3 mT , into the page
- (4) 3 mT , out of the page
- (5) 12 mT , into the page

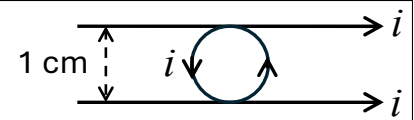


Solution: For circular motion in a magnetic field the angular frequency is

$$\frac{2\pi}{T} = \frac{qB}{m} \rightarrow B = \frac{m}{q} \frac{2\pi}{T}.$$

Using $T = 6\text{ ns}$, the magnetic field has magnitude 0.006 T or 6 mT . At the bottom of the path, the velocity is to the left and the force is upwards. For a negatively charged particle, this corresponds to a magnetic field **into the page**.

8. Two infinite wires and a circular loop each carry current $i = 3 \text{ A}$ in the directions indicated in the figure. The infinite wires are separated by 1 cm . What is the magnitude of the magnetic field at the center of the loop?



- (1) ♣ $3.8 \times 10^{-4} \text{ Tesla}$
 (2) $1.9 \times 10^{-4} \text{ Tesla}$
 (3) $6.2 \times 10^{-4} \text{ Tesla}$
 (4) $1.4 \times 10^{-4} \text{ Tesla}$
 (5) $5.0 \times 10^{-4} \text{ Tesla}$

Solution: The magnetic fields due to the infinite wires cancel at the center of the loop. This leaves only the magnetic field due to the loop, which has radius 0.5 cm . $B = \mu_0 i / (2r) = 3.8 \times 10^{-4} \text{ T}$.

9. The current through the inductor in a driven AC circuit is shown at right. Which of the following graphs corresponds to the voltage across the inductor?



Fig. 1

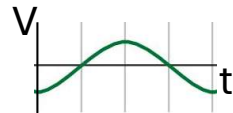


Fig. 2

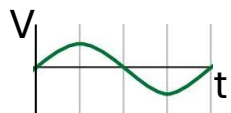


Fig. 3

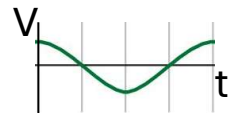


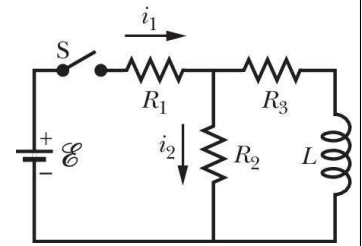
Fig. 4

- (1) ♣ Figure 1 (2) Figure 2 (3) Figure 3 (4) Figure 4 (5) none of the above

Solution: For an inductor the emf leads the current (ELI) by 90° or $\pi/2$ radians. Looking at the lowest point in the curves, in **Figure 1** the voltage leads the current by $\pi/2$ radians.

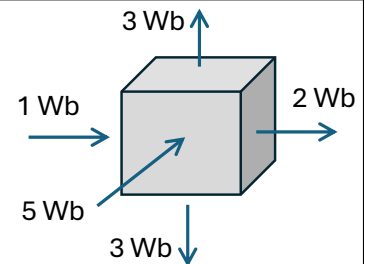
10. In the RL circuit at right each of the resistors has resistance 2Ω , and the inductor has inductance 5 mH . The battery has $\mathcal{E} = 12 \text{ V}$. What is the current i_2 right after the switch is closed?

- (1) ♣ 3 A
 (2) 1 A
 (3) 2 A
 (4) 4 A
 (5) 6 A



Solution: Right after the switch is closed, no current flows through the inductor. This means that $i_1 = i_2 = (12\text{V})/(2\Omega + 2\Omega) = 3 \text{ A}$.

11. The magnitude and direction of the fluxes through five of the faces of a cube are shown in the figure. What is the flux through the remaining face using the usual convention that positive flux is going outwards?



- (1) ♣ -2 Wb (2) 2 Wb (3) -15 Wb (4) 4 Wb (5) -4 Wb

Solution: Using the convention that the flux is positive going outwards, the net flux from the five surfaces indicated in the figure is $-5 + 3 + 3 + 2 - 1 = 2 \text{ Wb}$. Since the net magnetic flux through the entire surface is zero, the remaining surface must have a flux of -2 Wb .

12. The induced magnetic field at a radial distance 10 mm from the central axis of a circular parallel-plate capacitor is 4.0×10^{-7} Tesla. The plates have a radius of 5 mm. At what rate is the magnitude of the electric field between the plates changing in V/(m·s)?

- (1) ♣ 2.9×10^{13} (2) 0.7×10^{13} (3) 1.4×10^{13} (4) 3.5×10^{13} (5) 2.4×10^{13}

Solution: Since there is no charge transfer across the capacitor, the magnetic field is due to displacement current.

$$(4.0 \times 10^{-7} \text{ T}) 2\pi(10 \text{ mm}) = \oint \vec{B} \cdot d\vec{s} = \mu_o \epsilon_o \frac{d\Phi_E}{dt} = \mu_o \epsilon_o \pi(5 \text{ mm})^2 \frac{dE}{dt}$$

Solve for $dE/dt = \mathbf{2.9 \times 10^{13} \text{ V/(m} \cdot \text{s)}}$.

13. A laser beam of power 10 W and *diameter* 1 mm is directed upward and shines perpendicularly onto a flat face of a fully-reflecting cube of side a and density 2700 kg/m^3 . What is the largest value of a for which the force due to the laser light pressure can overcome the Earth's gravitational force? (Assume that the cube fits inside the laser beam; $g = 9.8 \text{ m/s}^2$.)

- (1) ♣ $3.2 \text{ } \mu\text{m}$ (2) $1.4 \text{ } \mu\text{m}$ (3) $25.6 \text{ } \mu\text{m}$ (4) $6.4 \text{ } \mu\text{m}$ (5) $12.8 \text{ } \mu\text{m}$

Solution: The intensity of the laser beam is $I = 10\text{ W}/(\pi r^2) = 1.27 \times 10^7 \text{ W/m}^2$, where $r = 0.0005\text{ m}$. The radiation pressure for reflection is $2I/c$, and the force on the block due to the light is $2Ia^2/c$, where a is the side of a the cube. The gravitational force is $\rho a^3 g$. Setting the two forces equal to each other implies that $a = 2I/(c\rho g) = 3.2 \times 10^{-6} \text{ m} = \mathbf{3.2 \text{ } \mu\text{m}}$.

14. Unpolarized light is sent along a z -axis through three consecutive polarizers placed in the (x, y) -plane. From the first to the last polarizers, their planes of polarization with respect to the x -axis are 90° , 30° , 60° . What fraction of the original light intensity gets through the three polarizers?

- (1) ♣ 9.4% (2) 42% (3) 38% (4) 28% (5) 19%

Solution: The first polarizer reduces the intensity of the incoming light by half because the incoming light is unpolarized. Each subsequent polarizer reduces the intensity by a factor of $\cos^2(\theta)$, where there θ is the angle between the polarizers. $I = 0.5I_o \cos^2(90 - 30) \cos^2(30 - 60) = 0.09375I_o$ or **9.4%**.

15. Which mirrors can form virtual images?

- (1) ♣ flat, concave, convex (2) flat and convex only (3) flat and concave only (4) convex only (5) concave only

Solution: If the object distance is positive, both flat and convex mirrors form virtual images. If the object distance is less than the focal length, a concave mirror forms a virtual image. Thus, **flat, concave, and convex** mirrors can form virtual images.

16. Two identical converging lenses each of focal length 1 m are placed 1 m apart. What is the overall magnification of the two-lens system for an object placed 50 cm away from the front lens?

- (1) ♣ -2 (2) -1 (3) $+1$ (4) $+2$ (5) $+4$

Solution: The first lens produces an image at

$$\frac{1}{0.5} + \frac{1}{i_1} = \frac{1}{1} \rightarrow i_1 = -1 \text{ m}.$$

Relative to the second lens this corresponds to an object being at $p_2 = 2 \text{ m}$. The image produced by the second lens is at

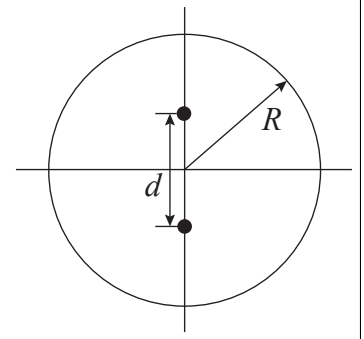
$$\frac{1}{2} + \frac{1}{i_2} = \frac{1}{1} \rightarrow i_2 = 2 \text{ m}.$$

The overall magnification of the two lens system is

$$M = \left(-\frac{i_1}{p_1}\right) \left(-\frac{i_2}{p_2}\right) = \mathbf{-2}.$$

17. Two coherent radio-frequency point sources separated by $d = 2.4$ m radiate in phase at a wavelength $\lambda = 1.2$ m. A radio wave detector moves in a circular path of some radius $R > d$ around the mid-point between the two sources (see figure) and measures the radio-wave intensity of the signal. Find how many minima does it detect.

- (1) ♣ 8
 (2) 6
 (3) 4
 (4) 12
 (5) 16



Solution: Where the circular path intersects the x-axis the path difference between the two sources is zero. Where the circular path intersects the y-axis the path difference between the two sources is d . This is the maximum path difference. Since $d/\lambda = 2$, there is constructive interference where the circular path intersects the y-axis. In between $\Delta L = 0$ and $\Delta L = 2\lambda$ there are two destructive interference minima. This is true for all four quadrants so there are a total of **8** minima.

18. Costume jewelry beads are made of glass with index of refraction 1.40. To make them more reflective, they are coated with a layer of material with index of refraction 1.60. What is the approximate minimum coating thickness needed for light of wavelength 550 nm and perpendicular incidence to be reflected with maximum constructive interference?

- (1) ♣ 86 nm (2) 138 nm (3) 98 nm (4) 172 nm (5) 69 nm

Solution: There is an additional phase difference of π when waves are reflected off a medium with a higher index of refraction. Thus, there is a phase shift of π for the coated layer because one reflects from $n = 1$ to $n = 1.6$. The additional phase shift of the light reflected from the glass bead is $2\pi(2d/1.6)$, where d is the thickness of the layer. Constructive interference occurs for

$$2\pi \frac{2d}{\lambda/1.6} - \pi = 2\pi m \rightarrow 2d = \frac{\lambda}{1.6} \left(m + \frac{1}{2}\right)$$

where m is an integer. The smallest positive d is for $m = 0$: **d = 86 nm**.

19. An amateur telescope has lens diameter 100 mm. What is the minimum separation between two objects on the moon that this telescope can resolve? Take the distance to the Moon to be 384,000 km, and the wavelength of the light to be 430 nanometers.

- (1) ♣ 2 km (2) 200 m (3) 2 m (4) 20 km (5) 20 m

Solution: Rayleigh's criteria is $\Delta\theta = 1.22\lambda/D = 1.22(430 \times 10^{-9})/(0.1) = 5.25 \times 10^{-6}$. This angle in radians is equal to the separation, s , between the objects divided by the distance to the Moon: $\Delta\theta = s/(3.84 \times 10^8 \text{ m})$ so the minimum resolvable separation is $s = 2000 \text{ m} = \mathbf{2 \text{ km}}$.

20. A diffraction grating has 400 lines per mm. Red light of wavelength 700 nanometers is incident perpendicular to the grating. How many constructive interference spots are seen on a screen 2 m away? This includes both the central spot, and the constructive interference spots to the right and left of it.

- (1) ♣ 7 (2) 3 (3) 4 (4) 5 (5) 6

Solution: The slits in the diffraction grating are separated by $d = 0.001\text{m}/400 = 2.5 \times 10^{-6}\text{m}$. The wavelength of the light is $\lambda = 7 \times 10^{-7}\text{m}$. Bright spots occur at $\sin(\theta) = m\lambda/d < 1$ so $m < d/\lambda = 3.57$. This means that m can take on the values $m = -3, -2, \dots, 2, 3$ or **7** possible values.

Scratch paper

Scratch paper

Scratch paper

Formulas – page 1

Constants: $e = 1.6 \times 10^{-19} \text{ C}$ $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$

$c = 3.0 \times 10^8 \text{ m/s}$ $k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$

$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$

kilo (k) = 10^3 ; centi (c) = 10^{-2} ; milli (m) = 10^{-3} ; micro (μ) = 10^{-6} ; nano (n) = 10^{-9} ; pico (p) = 10^{-12}

$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Coulomb's Law: $F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$

Electric field: $\vec{F} = q\vec{E}$

$E = \frac{Q}{r^2}k$ (point charge) $E = \frac{Q \cdot z}{(z^2 + R^2)^{3/2}}k$ (along a ring's axis)

$dE = \frac{dQ}{r^2}k$ (continuous) $E = 2 \frac{Q}{R^2}k \left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$ (along a disk's axis)

$E = \frac{Q}{\theta_0 R^2}k \cdot 2 \sin\left(\frac{\theta_0}{2}\right)$ (at arc center) $E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$ (infinite wire)

$E = \frac{\sigma}{2\epsilon_0}$ (infinite sheet)

Electric dipole:

Torque: $\vec{\tau} = \vec{p} \times \vec{E}$

Potential energy: $U = -\vec{p} \cdot \vec{E}$

$p = qd$

$\vec{p}$ direction: from negative to positive charge

Flux: $\Phi = \int \vec{E} \cdot d\vec{A} = \int E_{\perp} dA$, where $E_{\perp} = E \cos \theta$

$E = 0$ (inside conductor, any shape)

Gauss' Law: $\oint_{\text{closed surface}} \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$

$E = \frac{Q}{r^2}k$ (outside uniformly charged shell/sphere)

$E = 0$ (inside uniformly charged shell)

$E = \frac{Qr}{R^3}k$ (inside uniformly charged sphere of radius R)

$E = \frac{\sigma}{\epsilon_0}$ (next to surface of conducting material)

Electric potential:

$U = qV$ (potential energy)

$V = \frac{Q}{r}k$ (point charge)

$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$

$dV = \frac{dQ}{r}k$ (continuous)

$E_x = -\frac{\partial V}{\partial x}; \quad E_y = -\frac{\partial V}{\partial y}; \quad E_z = -\frac{\partial V}{\partial z}$

$U = \sum_{i \neq j} \frac{q_i q_j}{r_{ij}}k$ (potential energy of system of point charges)

Formulas – page 2

Capacitors: $q = CV$.

$$C_{\text{eq}} = \sum_{j=1}^n C_j \quad (n \text{ capacitors in parallel})$$

$$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^n \frac{1}{C_j} \quad (n \text{ capacitors in series}).$$

$$U_E = \frac{q^2}{2C} = \frac{1}{2} CV^2,$$

$$u_E = \frac{1}{2} \epsilon_0 E^2.$$

$$C = \frac{\epsilon_0 A}{d}. \quad (\text{parallel plate})$$

$$C = 2\pi\epsilon_0 \frac{L}{\ln(\frac{b}{a})}. \quad (\text{cylindrical})$$

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}. \quad (\text{spherical})$$

$$C = 4\pi\epsilon_0 R. \quad (\text{isolated sphere})$$

$$C = C_0 \cdot \kappa \quad (\text{with dielectric})$$

Current: $i = \frac{dq}{dt}, \quad i = \int \vec{J} \cdot d\vec{A}, \quad \vec{J} = (ne)\vec{v}_d,$

Resistors: $R = \frac{V}{i}, \quad \vec{J} = \frac{1}{\rho} \vec{E} = \sigma \vec{E}, \quad R = \rho \frac{L}{A}, \quad R = R_0 \cdot (1 + \alpha \cdot \Delta T) \quad P = iV = i^2 R = \frac{V^2}{R}.$

$$\frac{1}{R_{\text{eq}}} = \sum_{j=1}^n \frac{1}{R_j} \quad (n \text{ resistances in parallel}).$$

$$R_{\text{eq}} = \sum_{j=1}^n R_j \quad (n \text{ resistances in series}).$$

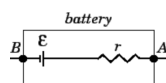
Battery:

$$\text{emf } \mathcal{E} = \frac{dW}{dq}$$

Ideal battery:

$$\text{---} \parallel \text{---} \quad \Delta V = \mathcal{E}$$

Realistic battery:



$$\Delta V = V_A - V_B$$

$$\Delta V \neq \mathcal{E}, \text{ when there is current}$$

Multi-loop circuits:

Preparatory work:

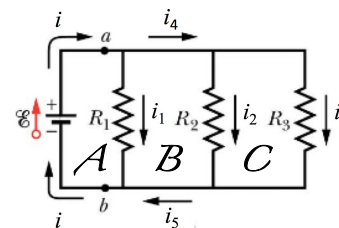
- Identify all independent loops A, B, C , etc. (each loop is like a boundary of a country)
- Mark currents i_1, i_2, i_3 , etc. for all wire segments between wire junctions (you can use arbitrary directions)

Kirchhoff's Rules :

- (1) Walk along each loop, full circle, and add differences of potentials as you jump over circuit elements (signs matter!). The sum for each loop is zero.
- (2) For each wire junction, the sum of incoming currents equals the sum of outgoing currents

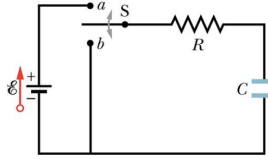
The obtained set of equations will allow you to find all unknowns (e.g. currents).

To find $V_a - V_b$, just follow any path from (a) to (b) and add differences of potentials as you jump over circuit elements.



Formulas – page 3

RC-circuits:



$$\tau = RC$$

Charging: (a) $q(t) = \mathcal{E}C(1 - e^{-t/\tau})$

Discharging: (b) $q(t) = \mathcal{E}Ce^{-t/\tau}$

$$i = \frac{dq}{dt}$$

Charged particles, currents in magnetic field:

Force on point charge: $\vec{F} = q\vec{v} \times \vec{B}$

Force on current segment: $\vec{F} = i\vec{L} \times \vec{B}$

Magnetic dipole:

- magnetic dipole of a coil with current $\vec{\mu} = iN\vec{A}$

- elementary particles have internal $\vec{\mu}$

- torque on a magnetic dipole $\vec{\tau} = \vec{\mu} \times \vec{B}$

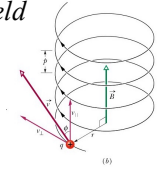
- potential energy of a magnetic dipole $U = -\vec{\mu} \cdot \vec{B}$

Motion of point charge in uniform B-field

- helical path

- $v_{\parallel}$ (velocity parallel to $\vec{B}$) = const

- in the plane perpendicular to $\vec{B}$, particle undergoes circular motion



$$R = \frac{mv_{\perp}}{qB} \quad T = 2\pi \frac{m}{qB} \text{ (independent of velocity!)}$$

Motion of point charge crossed E and B:

If $\vec{E} \perp \vec{B}$, $v = \frac{E}{B}$, and $\vec{v} \parallel (\vec{E} \times \vec{B})$,

then, the net force on a charged particle is zero

Hall effect: $V = vBw$ $V = \frac{iB}{\text{end}}$
 w – width d – thickness

Magnetic field due to moving charges, currents:

- moving point charges: $\vec{B} = \left(\frac{\mu_0}{4\pi}\right) q \frac{\vec{v} \times \hat{r}}{r^2}$,

- segments of current: $d\vec{B} = \left(\frac{\mu_0}{4\pi}\right) i \frac{d\vec{s} \times \hat{r}}{r^2}$

- infinitely long wire: $B = \frac{\mu_0 i}{2\pi R}$

- arc at the circle center point $B = \frac{\mu_0 i}{4\pi R} \phi$

- loop along the z-axis: $B = \frac{\mu_0 i}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$

- magnetic dipole ($\vec{\mu}$) along its axis $\vec{B} = \frac{\mu_0 \vec{\mu}}{2\pi z^3}$

Ampere's Law: $\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{encircled}}$

Solenoid:

Inside $B = \mu_0 i n$

outside $B = 0$



Toroid:

Inside $B = \mu_0 i \frac{N}{2\pi r}$

outside $B = 0$



Force between two parallel wires

Same direction currents: attraction $F = \frac{\mu_0 L}{2\pi d} i_a i_b$

Opposite direction currents: repulsion

Induced electric field, emf:

Faraday's law: $\mathcal{E} = -\frac{d\Phi_B}{dt}$, where

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s} \quad \Phi_B = \int \vec{B} \cdot d\vec{A}$$

EMF direction: three-step right-hand rule

Lenz's law: if there is a conductive loop, induced EMF will give rise to current. The consequences from this induced current (B-field, forces/torque on the loop) will appear to oppose the change in the original flux

Inductor $L = \frac{N\Phi_B}{i} \quad \mathcal{E}_L = -L \frac{di}{dt}$

Inductance of solenoid $L = \mu_0 n^2 A \ell$

Mutual inductance $\mathcal{E}_2 = -M \frac{di_1}{dt}$

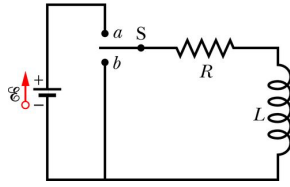
of a pair of inductors $\mathcal{E}_1 = -M \frac{di_2}{dt}$

Energy stored in inductor $U = \frac{1}{2} Li^2$

Magnetic field energy density $u_B = \frac{B^2}{2\mu_0}$

Formulas – page 4

RL-circuits:



$$\tau = \frac{L}{R}$$

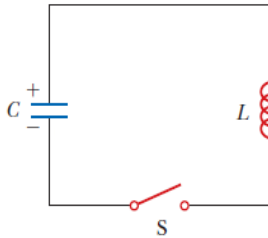
$$(a) \quad i(t) = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau})$$

$$(b) \quad i(t) = \frac{\mathcal{E}}{R} e^{-t/\tau}$$

In circuits with inductors (no driving AC emf):

- After a long time, all currents in a circuit are steady and an inductor can be thought of as a mere conductive wire (no induced emf across it)
- At the instance right after a switch is flipped, current in inductor cannot change abruptly (i.e. it is the same as just before the switch has been flipped)

LC and RLC circuits (without driving emf)

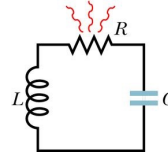


$$\omega = \frac{1}{\sqrt{LC}}$$

$$q(t) = Q_{\max} \cos(\omega t + \phi)$$

$$i(t) = q'(t) = -\omega Q_{\max} \sin(\omega t + \phi)$$

$$U = \frac{1}{2} \frac{q^2(t)}{C} + \frac{1}{2} L i^2(t) = \text{const}$$

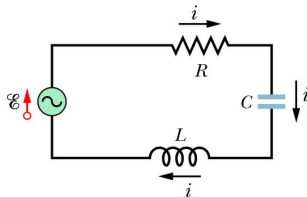


$$q(t) = Q_{\max} e^{-t/\tau} \cos(\omega' t + \phi)$$

$$i(t) = -\omega Q_{\max} e^{-t/\tau} \sin(\omega' t + \phi + \alpha), \text{ where } \sin \alpha = \frac{1}{\omega \tau}$$

$$\tau = \frac{2L}{R}, \quad \omega' = \sqrt{\omega^2 - \left(\frac{1}{\tau}\right)^2}$$

LC and RLC circuits (with driving AC emf)



$$\mathcal{E}(t) = \mathcal{E}_{\max} \sin(\omega_d t)$$

$$i(t) = \frac{\mathcal{E}_{\max}}{Z} \sin(\omega_d t - \phi)$$

$$\mathcal{E}_{\text{rms}} = \frac{\mathcal{E}_{\max}}{\sqrt{2}}, \quad i_{\text{rms}} = \frac{i_{\max}}{\sqrt{2}}, \quad i_{\text{rms}} = \frac{\mathcal{E}_{\text{rms}}}{Z}$$

$$\text{Average power provided by emf: } \langle P(t) \rangle = \mathcal{E}_{\text{rms}} i_{\text{rms}} \cos \phi$$

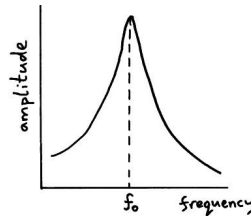
$$\text{Average power dissipated on R: } \langle P(t) \rangle = i_{\text{rms}}^2 R$$

$$\mathcal{E} - R \text{ circuit: } Z = R, \quad \phi = 0$$

$$\mathcal{E} - C \text{ circuit: } Z = X_C = \frac{1}{\omega_d C}, \quad \phi = -\frac{\pi}{2}$$

$$\mathcal{E} - L \text{ circuit: } Z = X_L = \omega_d L, \quad \phi = +\frac{\pi}{2}$$

$$\mathcal{E} - RLC \text{ circuit: } Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan \phi = \frac{X_L - X_C}{R}$$



$$\text{Amplitude } i_{\max} = \frac{\mathcal{E}_{\max}}{Z(\omega)}$$

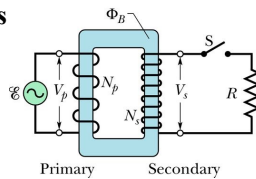
$$\text{Resonance is when } \omega_d = \omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow Z_{\min} = R$$

$$\Rightarrow \text{Current amplitude is max possible}$$

$$\Rightarrow \phi = 0$$

Transformers



$$\mathcal{E}_{\text{secondary}} = \frac{N_s}{N_p} \mathcal{E}_{\text{primary}}$$

$$i_{\text{secondary}} = \frac{N_p}{N_s} i_{\text{primary}}$$

$$P_{\text{secondary}} [\text{consumed}] = P_{\text{primary}} [\text{provided}]$$

$$R_{\text{eq}} = \left(\frac{N_p}{N_s}\right)^2 R$$

Formulas – page 5

Gauss Law for B, induced B due to dE/dt, magnetic materials

$$\oint \vec{B} \cdot d\vec{A} = 0$$

This implies there are no magnetic monopoles

$$\oint \vec{B} \cdot d\vec{l} = + \mu_0 \epsilon_0 \cdot \frac{d\Phi_E}{dt}$$

USE RIGHT HAND RULE to get the direction of induced field

Magnetic properties of materials: $B = \mu B_{\text{vacuum}}$

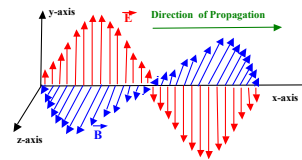
- Paramagnetic materials: $\mu > 1$
- Diamagnetic materials: $\mu < 1$
- Ferromagnetic materials: $\mu \gg 1$ (hysteresis, critical temperature)
- Superconductors: $B=0$ (exactly)

EM waves

General equation for a harmonic wave $A(x,t) = A_0 \sin(kx \mp \omega t)$

- wave propagates in $\pm x$ direction
- wavelength $\lambda = 2\pi / k$
- period $T = 2\pi / \omega$
- frequency $f = 1/T = \omega / (2\pi)$
- velocity $v = \omega / k = \lambda / T$

$$A_{rms} = \frac{A_0}{\sqrt{2}}$$



$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 299,792,458 \text{ m/s}$$

ALWAYS:

$$B = \frac{E}{c}$$

$$\vec{E} \perp \vec{B}$$

Direction: $\vec{E} \times \vec{B}$ (right-hand rule)

$$\text{EM wave energy flow (J per s per m}^2\text{)} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$\text{Intensity (energy flow averaged over time)} \langle |\vec{S}| \rangle = \frac{1}{\mu_0} E_{rms} B_{rms} = \frac{1}{c\mu_0} E_{rms}^2 = \frac{c}{\mu_0} B_{rms}^2$$

$$\text{Intensity distance } r \text{ away from a point source of power } P_s \text{ (Watt)} \quad I = \frac{P_s}{4\pi r^2}$$

Pressure (N/m²) on

$$100\% \text{ absorbing surface: } P = \frac{1}{c} I$$

$$100\% \text{ reflective surface: } P = \frac{2}{c} I$$

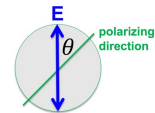
EM wave in medium with index of refraction n :

$$v_n = c / n, \quad f_n = f, \quad \lambda_n = \lambda / n$$

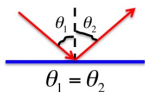
Fraction of light intensity coming through a polarizing sheet:

Unpolarized light: $\frac{1}{2}$

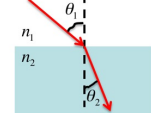
Polarized light: $\cos^2(\theta)$



Reflection:

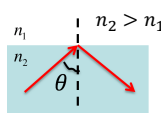


Refraction:



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Total reflection:



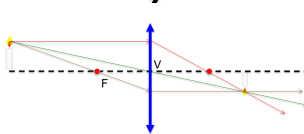
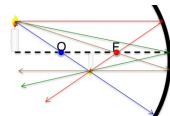
$$\theta > \theta_c, \text{ where } \sin \theta_c = \frac{n_1}{n_2}$$

Mirrors, lenses, optical devices

$$\text{Concave/convex mirror: } f = \pm \frac{R}{2}$$

$$\text{Lens: } \frac{1}{f} = (n-1) \left(\pm \frac{1}{R_1} \pm \frac{1}{R_2} \right),$$

where +/- for convex/concave surface as seen from outside of the lens



$$\frac{1}{f} = \frac{1}{p} + \frac{1}{i} \quad \text{magnification } m = -\frac{i}{p}$$

$i = +/-$ (real/virtual image)
 $m = +/-$ (upright/inverted image)

Correction lenses:

$$D = \frac{1}{f} = \frac{1}{\infty} + \frac{1}{-d_{\text{fartherst}}}$$

$$D = \frac{1}{f} = \frac{1}{0.25\text{m}} + \frac{1}{-d_{\text{closest}}}$$

Typical minimum distance at which one can see objects clearly: $p_{\min} = 25 \text{ cm}$

Magnifying glass:

$$m = \frac{p_{\min}}{f}$$

Microscope:

$$m = -\frac{d}{f_{\text{objective}}} \cdot \frac{p_{\min}}{f_{\text{eyepiece}}}$$

Telescope:

$$m = -\frac{f_{\text{objective}}}{f_{\text{eyepiece}}}$$

Formulas – page 6

Interference

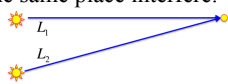
Two coherent waves arriving to the same place interfere.

$$A_1 = A \sin(kL_1 - \omega t) = A \sin(\Phi_1)$$

$$A_2 = A \sin(kL_2 - \omega t) = A \sin(\Phi_2)$$

Constructive interference: $\Delta\Phi = \Phi_2 - \Phi_1 = 2\pi m$, $m = 0, \pm 1, \pm 2, \dots$

Destructive interference: $\Delta\Phi = \Phi_2 - \Phi_1 = \pi + 2\pi m$



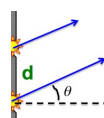
Two slits:

$$\Delta\Phi = k(d \sin \theta) = \frac{2\pi}{\lambda} d \sin \theta$$

$$I = 4 \cos^2 \left(\pi \frac{d \sin \theta}{\lambda} \right) \cdot I_0$$

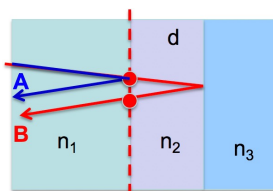
$$\text{max: } d \sin \theta_{\text{max}} = m\lambda$$

$$\text{min: } d \sin \theta_{\text{min}} = \left(m + \frac{1}{2} \right) \lambda$$



Refraction does not change EM wave phase on the boundary between two media

Reflection changes EM wave phase by π ONLY IF the wave reflects off medium with higher index of refraction



$$\Phi_A = \pi (0), \text{ for } n_2 > n_1 \text{ (} n_2 < n_1 \text{)}$$

$$\Phi_B = \frac{2\pi}{\lambda_{n_2}} d + \pi (0) + \frac{2\pi}{\lambda_{n_2}} d, \text{ for } n_3 > n_2 \text{ (} n_3 < n_2 \text{)}$$

Diffraction

Single slit diffraction (slit width a):

$$\text{Directions toward minima: } a \sin \theta_m^{\text{min}} = m\lambda$$

Diffraction on circular aperture (diameter D)

$$\text{Telescope angular resolution } \Delta\theta \approx 1.22 \frac{\lambda}{D}$$

Diffraction grating (N slits spaced by distance d apart)

$$\text{Directions toward maxima: } d \sin \theta_m^{\text{max}} = m\lambda$$

$$\text{Two wavelength can be resolved, if } \frac{\Delta\lambda}{\lambda} > \frac{1}{Nm}$$

Diffraction on crystals (separation between atom layers – d)

$$\text{Directions toward maxima: } 2d \sin \theta_m^{\text{max}} = m\lambda$$

Details of Master Exam Questions

Total of 20 QQ, 0 CQ, 0 RQ questions (0 distinct RQ questions)

- Total of 20 questions defined (20 for each student exam)
- Total weight is 20.00 for each student exam
- Total of 18 figures defined

Q	Type	aType	Wght	#fig	#ans	#ran	Correct	Short text
1	QQ	Single	1.00	0	5	5	1	Three point charges of $200\ \mu\text{C}$ each are positioned i
2	QQ	Single	1.00	1	5	5	1	Which of the configurations with four charges placed at
3	QQ	Single	1.00	1	5	5	1	A small ball, with a charge of $+2\ \mu\text{C}$ on it, is pl
4	QQ	Single	1.00	1	5	5	1	In the capacitor circuit in the figure all the capacito
5	QQ	Single	1.00	1	5	5	1	In the figure the voltage as a function of position is
6	QQ	Single	1.00	1	5	5	1	1. In the resistor circuit shown in the figure each of
7	QQ	Single	1.00	1	5	5	1	An electron enters a region with a uniform magnetic fie
8	QQ	Single	1.00	1	5	5	1	Two infinite wires and a circular loop each carry curre
9	QQ	Single	1.00	2	5	5	1	The current through the inductor in a driven AC circuit
10	QQ	Single	1.00	1	5	5	1	In the RL circuit at right each of the resistors has re
11	QQ	Single	1.00	1	5	5	1	The magnitude and direction of the fluxes through five
12	QQ	Single	1.00	0	5	5	1	The induced magnetic field at a radial distance 10 mm f
13	QQ	Single	1.00	0	5	5	1	A laser beam of power 10 W and <i>diameter</i> 1 mm is d
14	QQ	Single	1.00	0	5	5	1	Unpolarized light is sent along a z -axis through thre
15	QQ	Single	1.00	0	5	5	1	Which mirrors can form virtual images?
16	QQ	Single	1.00	0	5	5	1	Two identical converging lenses each of focal length 1
17	QQ	Single	1.00	1	5	5	1	Two coherent radio-frequency point sources separated by
18	QQ	Single	1.00	0	5	5	1	Costume jewelry beads are made of glass with index of r
19	QQ	Single	1.00	0	5	5	1	An amateur telescope has lens diameter 100 mm. What is
20	QQ	Single	1.00	0	5	5	1	A diffraction grating has 400 lines per mm. Red light o

Explanation of columns

- Type = question type (QQ, CQ, RQ)
- aType = answer type (Single choice, True/False)
- Wght = question weight (typically 1.0)
- #fig = number of figures
- #ans = number of answers
- #ran = number of randomized answers (0 $\Rightarrow$ answers not randomized)
- Correct = List of correct answers